ENTERPRISE ARCHITECTURE SPEC

Project 1: "OmniBrain"

Agentic Multi-Modal Retrieval-Augmented Generation (RAG) Orchestrator

DOMAIN
Large Language Models & Agentic AI

CORE FRAMEWORKS
LangGraph, Qdrant/FAISS, Fast API, Streamlit

TARGET DEPLOYMENT
Enterprise Financial & Analytics Platform

1. Project Overview & Problem Statement

Standard Retrieval-Augmented Generation (RAG) architectures encounter severe limitations when applied to enterprise document analysis. Standard systems process text sequentially and struggle with complex document layouts containing multimodal elements—such as quantitative financial tables, embedded charts, vector graphics, and scanned figures. Furthermore, single-step retrieval fails when complex queries require multi-step analytical reasoning across disparate data sources.

> **The Enterprise Problem:** Standard vector retrieval models flatten document structures into text chunks, completely losing tabular cell dependencies and embedded visual insights. When answering complex queries (e.g., comparing historical stock metrics in a database against future growth forecasts in a 500-page corporate PDF), traditional RAG frequently hallucinates or produces incomplete analytical responses.

OmniBrain Solution: OmniBrain introduces an enterprise-grade, agentic multi-modal orchestration system. Powered by **LangGraph**, OmniBrain dynamically routes sub-tasks across specialized AI agents. It uses **Vision-Language Models (VLMs)** to parse visual charts and structured tables, semantic vector search (Qdrant/FAISS with CLIP and text embeddings) for unstructured context, and a dedicated **Text-to-SQL Agent** to query relational databases.

2. Enterprise Use Case: Quantitative Financial Memo

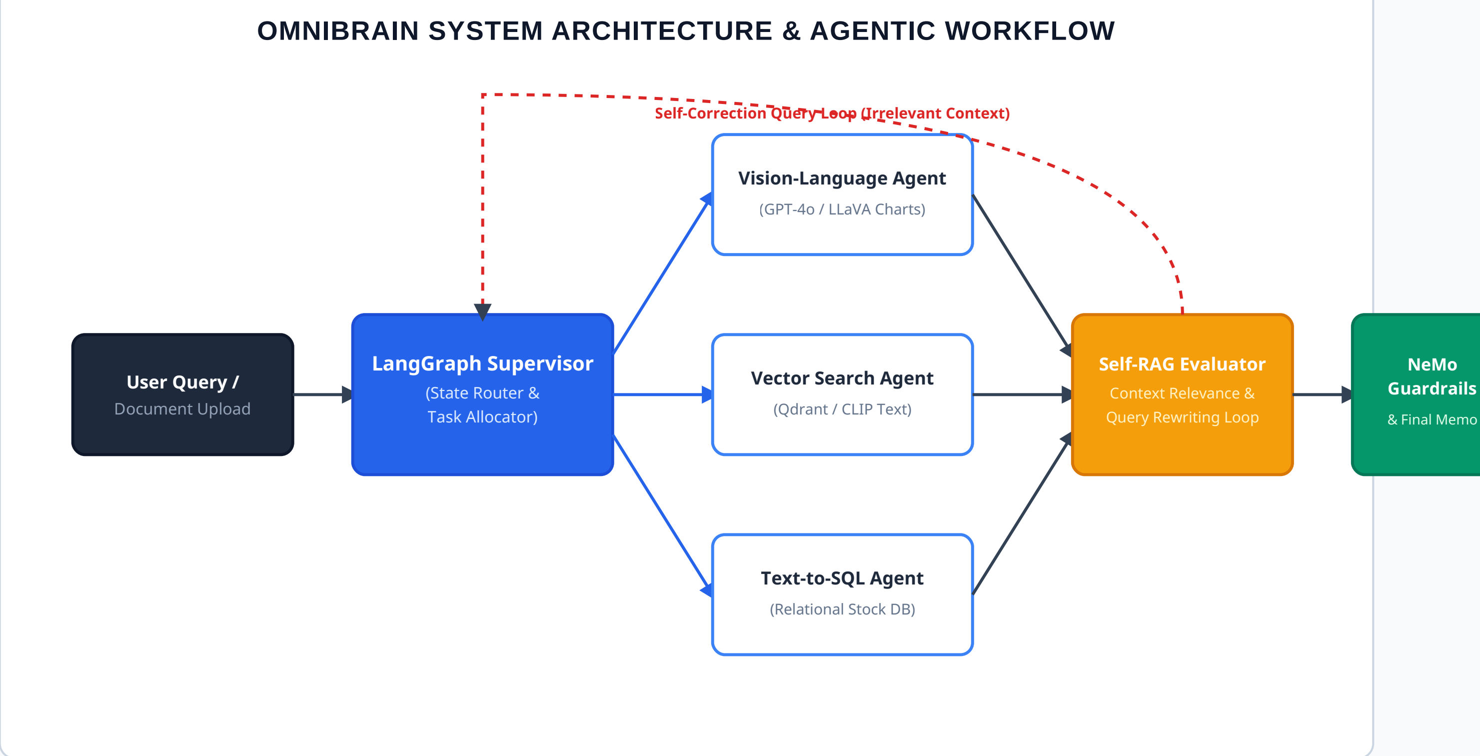
Consider a financial analyst evaluating a 500-page corporate financial PDF. A query such as: *"Cross-reference Q3 revenue growth shown in the corporate report's bar charts against historical stock performance in our SQL database, and evaluate risks."* requires multi-agent coordination:

- **Document Parsing:** The VLM Agent extracts exact numerical coordinates and metrics from embedded bar charts and financial tables without losing layout structure.
- **Relational DB Integration:** The SQL Agent generates verified database queries to extract historical stock price trends and financial ratios from relational storage.
- **Semantic Knowledge Synthesis:** The Vector Search Agent retrieves semantic textual insights regarding qualitative business risk factors.

- **Self-Correction & Guardrails:** The LangGraph orchestrator cross-checks retrieved context for accuracy, self-corrects search queries if retrieval confidence is low, enforces NeMo Guardrails, and synthesizes a fully cited, hallucination-resistant investment memo.

3. System Architecture & Agent Workflow

The diagram below illustrates the end-to-end multi-agent orchestration pipeline, from multimodal ingestion to dynamic query routing, self-correction, guardrail checks, and cited generation.



4. Technical Specifications & Core Modules

1. LangGraph Agentic Orchestrator

Serves as the cognitive state engine. Manages dynamic routing between tools based on state graph transitions. Implements explicit state memory to handle multi-turn user queries without context loss.

3. Vision-Language Model (VLM)

Utilizes GPT-4o / LLaVA to process extracted document visual elements. Converts line charts, scatter plots, and complex multi-column tables directly into structured Markdown tables or JSON formats.

2. Multi-Modal Retrieval System

Integrates Qdrant / FAISS for dual-vector indexing. Converts text into dense vectors via standard embeddings, while extracting and vectorizing embedded document visual assets using CLIP embeddings.

4. Guardrails & Observability

Monitored via **Langfuse** for token usage tracking, response latency, and complete execution path tracing. Enforces **NeMo Guardrails** to prevent off-topic or hallucinated queries.

5. Self-Correction Mechanism (Self-RAG)

To eliminate hallucination risks, OmniBrain employs a formal **Self-RAG framework**. Retrieved document chunks $D = \{d_1, d_2, ..., d_k\}$ for a user prompt q are scored for relevance using a critique model $C_{rel}(q, d_i)$:

$$Score = \sum_{i=1}^{k} C_{rel}(q, d_i) \cdot sim(E(q), E(d_i))$$

If $Score < \tau$ (where threshold $\tau = 0.72$), the orchestrator flags low retrieval quality, rejects the retrieved set, triggers an internal query rewrite module $q' = R(q)$, and re-queries the vector database before generating any user-facing output.

6. Week-Wise Development Plan & Milestones

Week	AI Engineering (LangGraph, VLMs, Vector DB)	Full-Stack & System Integration (FastAPI, Streamlit)
Week 1	**Multi-Modal Ingestion:** Build PDF parsing pipeline, chunk unstructured text, extract visual charts/tables. Generate dense embeddings using CLIP and store in Qdrant vector database.	**API Scaffolding:** Develop asynchronous FastAPI backend endpoints for document uploading, background document processing, and query ingestion.
Week 2	**Agentic Architecture:** Implement LangGraph state machine. Build "Supervisor" routing node to classify incoming user queries and delegate tasks to sub-agents.	**Chat UI Implementation:** Construct interactive Streamlit dashboard. Display agent execution state, thought process tree, and retrieved source images.
Mid Review	**Milestone Audit:** Prove Supervisor node correctly routes queries (Vector Search vs. SQL vs. VLM). Verify VLM accuracy in extracting exact numerical data from bar chart image queries.	
Week 3	**Self-Correction Loop:** Implement Self-RAG logic. Integrate self-critique nodes to detect low-relevance context and automatically trigger query rewrite cycles.	**Guardrails Enforcement:** Integrate NeMo Guardrails to restrict response generation strictly within the domain of provided corporate documents.
Week 4	**Evaluation Metrics:** Connect Langfuse tracking to monitor token costs, LLM execution latency, and agent trace graphs for complete system observability.	**Refine & Polish:** Complete UI polish with interactive citation links. Allow users to click an AI-generated claim to open the exact PDF page and source visual chart.
Final Review	**Production Readiness:** Enterprise-grade Agentic AI platform capable of autonomous reasoning, visual chart analytical execution, and hallucination-resistant search.	